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Investigating capabilities of Food Hubs in short food supply chains

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ABSTRACT

Food Hubs (FHs) have been presented as a possible intermediary increasing competitive advantage of SFSCs. However, research on the capabilities of FHs related to sustainability indicators is limited. This study aims to investigate the capabilities of FHs and its relation to sustainability by conducting explorative case studies in Sweden. Six FHs have been investigated through semi-structured interviews, observations, documentation and archival records. Their capabilities have been presented in this study categorized in operations management, producer support and community engagement. Findings show that the FHs operational capabilities are close to conventional intermediaries, while their relational capabilities are unique resources contribution to competitive advantage. With a focus on economic and environmental sustainability in their capabilities, FH's capabilities related to social sustainability such as community engagement is less present. Given the role of community engagement in achieving economies of scale, FHs should strengthen these relational capabilities to enhance their competitive advantage within SFSCs. The paper concludes by outlining directions for future research.

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KEYWORDS
Food hub; SFSC; sustainability; capabilities; competitive advantage

1. Introduction

Currently, most food products are sold through conventional food supply chains (CFSCs) where intermediaries such as wholesalers and retailers are responsible for the highest national food turnover (Manikas, Malindretos, and Moschuris 2019). To create a more sustainable food supply chain, scholars direct their attention to the development of short food supply chains (SFSCs). SFSCs are localized systems that employ direct or short-intermediate distribution and marketing channels. In contrast to CFSCs, they operate within defined geographical and social contexts, prioritizing sustainability, transparency, and community-oriented food production and consumption (Charatsari, Lioutas, and De Rosa 2024; Mustapa and Kallas 2025). While direct-to-consumer SFSCs are often discussed in the literature as a sustainable solution, they do not consider the necessary capabilities intermediaries fulfill with regards to processing, sales and distribution, among other, for small and medium-sized producers (Renkema and Hilletofth 2022).

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As a result, intermediate SFSCs are developed at regional levels that involve more intermediaries, but have a large focus on collaboration and sustainability in the supply chain (SC), and exist as a values-based SCs (Chiffoleau and Dourian 2020). However, these SCs are more often dominated by intermediaries such as conventional wholesalers and retailers, selling the largest volume of regional products, creating new power imbalances and increasing the risk for unfair trading practices (Malak-Rawlikowska et al. 2019; Chiffoleau and Dourian 2020). Rather than upholding the collaborative relationship-driven philosophy connected to SFSCs, these intermediaries often operate in a demand-driven SC, where the customer demands and financial profits are often prioritized over relationships with producers (Bloom and Hinrichs 2011; Chiffoleau and Dourian 2020).

To increase competitive advantage of SFSCs towards CFSCs, while at the same time focusing on sustainability and shared value amongst all stakeholders, there is a demand for a new type of intermediary. An intermediary that is profit-driven and competing with the larger conventional actors, but motivated to do so while operating as an enabler for producers, increasing the economic and social conditions of the latter, ultimately enhancing local and regional markets (Brunori et al. 2016). One possible intermediary that could respond to the barriers faced by small and medium sized actors in SFSCs is the Food Hub (FH) (Hyland and Macken-Walsh 2022). FHs have been defined as a value-based and mission-driven business that assembles or aggregates local or regional products with the goal to distribute these either by the development of own channels, or by using existing channels, to consumers as well as business customers, including public procurement institutions (Anggraeni, Handayati, and Novani 2022; Artoonian, Ross, and Shupp 2023; Barham et al. 2012; Berti and Mulligan 2016; Matson and Thayer 2013).

So far, research on this specific intermediary is limited. Most recent research quote Barham et al.'s (2012) report, as well as Berti and Mulligan's (2016) paper. These papers describe the definitions of FHs in the literature and give account of the different capabilities FHs have in the SC. This research, however, does not discuss the capabilities of FHs in relation to competitive advantage in SFSCs, neither does this research link the capabilities to specific sustainability indicators. Moreover, the concepts are described within the geographical limitations of the U.S.A. market, with the concept of FHs in Europe being underexplored. Additionally, in the recent years, many crises have revealed the weaknesses in the existing food SCs, where the COVID19 pandemic caused the closing of borders, and the Russia/Ukraine war caused a surge in price for agricultural seeds and fertilizers causing large problems for small- and medium-sized producers (Remvig and Hansson 2022).

There is some knowledge regarding the intermediaries that have shown greater capacity to move larger volumes in SFSCs (Malak-Rawlikowska et al. 2019), but little is known about the FHs resources, and capabilities that enable this success. Addressing this gap is essential to understand how the FH can drive the long-term growth and sustainability of SFSCs. Therefore, this paper aims to investigate the capabilities of FHs contributing to competitive advantage in SFSCs and linking them to sustainability indicators.

The research questions that this paper aims to answer are:

- What are the capabilities of Food Hubs in SFSCs?
- How are they related to sustainability?

Empirical data is collected through an explorative multiple case study into six different FHs in Sweden. The Swedish context is interesting due to its geographical location in the north of Europe, with different climate zones and prerequisites for agriculture, where the physical distance between producers and consumers is large and many inhabitants living in cities.

The outcome of this paper will allow for the assessment of capabilities that could strengthen the FH's operations, ultimately leading to competitive advantage. This paper contributes to the existing literature on SFSC and FHs, by providing knowledge regarding FHs capabilities. The paper also extends the existing literature by examining these capabilities in the context of the resource-based view (RBV).

This paper starts with a literature review of the available information with regards to the definition and the capabilities of FHs and explains the RBV lens that is used. In the methods section, the authors report the method of the explorative multiple case study into six different FHs in Sweden. Where the companies were studied holistically and compared by means of a cross-case analysis. Results are reported in section 4 and are compared with capabilities reported in earlier studies by Berti and Mulligan (2016) and Barham et al. (2012). The capabilities of the different companies are being compared with each other in the discussion section, answering the research questions. Finally, the conclusion section will provide recommendations for future research.

2. Literature review

2.1. The resource-based view

The RBV is a strategic management theory emphasizing a firm's internal resources and capabilities as sources of competitive advantage (Barney 1991). These include tangible assets and intangible resources such as knowledge, processes, and organizational capabilities. Sustained competitive advantage arises when resources are valuable, rare, inimitable, and non-substitutable (VRIN). The RBV shifts focus from external market conditions to leveraging unique internal strengths, guiding firms to identify and nurture VRIN resources for long-term performance (Christopher 2023). In a SC context, relationships with suppliers, partners, and customers can be strategic resources when VRIN. Collaborative capabilities, such as coordination, communication, and network-building, enhance SC integration, agility, and value delivery, which are difficult for competitors to replicate (Kim, Song, and Triche 2015). This emphasis on inter-firm collaboration is particularly relevant for SFSCs, where socially complex and ambiguous resources are derived from SC networks and create unique advantages that competitors cannot easily imitate, ensuring sustained competitive performance (Gold, Seuring, and Beske 2010).

2.2. Competitive advantage and sustainability in SFSCs

In recent years, scholars have highlighted SFSCs as sustainable alternatives to CFSCs. SFSCs are argued to stimulate local and regional economies (Deller, Lamie, and Stickel 2017), while addressing environmental, health, cultural, and animal welfare concerns (Bosona and Gebresenbet 2011). Increased agricultural activity in rural areas benefits not only producers but society, contributing to community resilience and regional

development (Brunori et al. 2016; Galli and Brunori 2013). Economically, SFSCs allow producers to capture higher profit margins by reducing intermediaries (Malak-Rawlikowska et al. 2019; Vittersø et al. 2019), while consumers benefit from fresher produce and reduced food miles, lowering CO_2 emissions (Nordmark et al. 2012). Beyond economic and environmental gains, SFSCs foster social capital by strengthening ties among producers, consumers, and other stakeholders, reinforcing local identity and cooperation (Galli and Brunori 2013). These advantages stem not only from operational resources, such as logistics and marketing, but also from relational capabilities, trust, collaboration, and community engagement, that are essential in SFSCs, particularly for small and medium producers in rural markets (Barney 1991; Christopher 2023; Mejia and Garcia-Diaz 2018). Such socially embedded resources have the possibility of enhancing SFSCs' ability to coordinate transactions and differentiate themselves from conventional actors (Schreiber et al. 2023).

Yet, despite these potential advantages, SFSCs face significant challenges. Their environmental performance is not inherently superior as their production methods and CO_2 emissions may be comparable to CFSCs (Brunori et al. 2016). Especially in the case of last-mile logistics (from the retailer to the consumer), recent studies further reveal mixed results in environmental logistics, while SFSC outperform CFSC retail in upstream logistics, downstream impacts are often worse due to inefficient consumer travel patterns (Falasco et al. 2025). Moreover, SFSCs often struggle with accessibility and inclusivity, as local food is frequently priced higher than conventional alternatives, it limits participation by lower-income groups and risks reinforcement of inequalities (Allen et al. 2003). This 'local food trap' (Born and Purcell 2006) highlights how outcomes depend less on scale than on underlying power relations (Brunori et al. 2016; Landon and Rosol 2025). As a result, SFSCs may inadvertently privilege middle-class consumers with the cultural and financial capital to engage (Hinrichs 2003), while excluding marginalized populations (Sage 2003).

Scaling up also remains difficult and despite rising interest, SFSCs' volumes are still limited, constraining competitiveness with CFSCs (Malak-Rawlikowska et al. 2019). The long-term competitiveness of SFSCs will depend on their ability to develop and sustain their internal and external resources, not only in operations, but in socially complex relational capabilities such as producer support, consumer trust, and community engagement, that are difficult for CFSC to replicate (Barney 1991; Gold, Seuring, and Beske 2010).

Finally, measuring sustainability outcomes remains a challenge. Many indicators are qualitative or self-reported, making it difficult to assess concrete impacts. Scholars stress the need for 'hard' quantitative methods, such as life-cycle assessments, complemented by 'soft' approaches that capture trade-offs and support reflexive governance of food systems (Brunori et al. 2016).

2.3. Defining Food Hubs

One intermediary having the potential to increase competitive advantage in SFSC is the FH. FHs are intermediaries that aggregate and sell local or regional products while creating value for multiple stakeholders, acting as coordinators between producers and customers and aligning common interests in food systems (Hyland and Macken-Walsh 2022). Unlike conventional intermediaries, FHs operate under both business and mission-driven objectives, combining conventional strategies with mission-driven strategies

related to community development and sustainability (Tewari et al. 2018; Trivette 2019). Relationships with stakeholders are important and producers are treated as strategic partners, rather than anonymous suppliers (Dimitri and Gardner 2019; Stevenson et al. 2011).

Although often framed as profit-oriented enterprises, some FHs operate under alternative models. LeBlanc et al. (2014) describe non-profit FHs supported by government initiatives, while Rogers and Fraszczak (2014) analyze a producer-led hub where farmers co-owned the platform, using it to foster horizontal collaboration, coordinate supply, and access larger buyers. Such models emphasize trust and collective action but face challenges in competing with conventional actors.

FHs vary in market orientation. Direct-to-consumer FHs, function as retailers, offering box schemes and diverse assortments while emphasizing logistics, marketing, collaboration, and community engagement, capabilities linked to achieving both competitive advantage and economic sustainability (Kao et al. 2024). Producer-to-business FHs, by contrast, supply larger customers such as HoReCa and public procurement, while hybrid FHs target both consumer and business markets (see Figure 1) (Dimitri and Gardner 2019; Hyland and Macken-Walsh 2022).

Differences also exist in strategic orientation. Manikas, Malindretos, and Moschuris (2019) distinguish between value-based FHs, emphasizing objectives such as scaling up local food systems, enhancing profitability, expanding distribution channels, and addressing power imbalances within the SC, and sustainability-oriented FHs, which prioritize environmental protection, the social economy and societal well-being. Similarly, Tewari et al. (2018) view FHs as mission-driven but competitive intermediaries, generating sufficient returns while maintaining long-term commitments to local producers. These FHs mobilize both tangible resources (e.g. logistics, infrastructure) and intangible capabilities (trust, coordination, commitment), thereby enhancing producers' bargaining power and access to markets (Manikas, Malindretos, and Moschuris 2019). Beyond aggregation and sales, many FHs provide branding, packaging, processing, and coordination

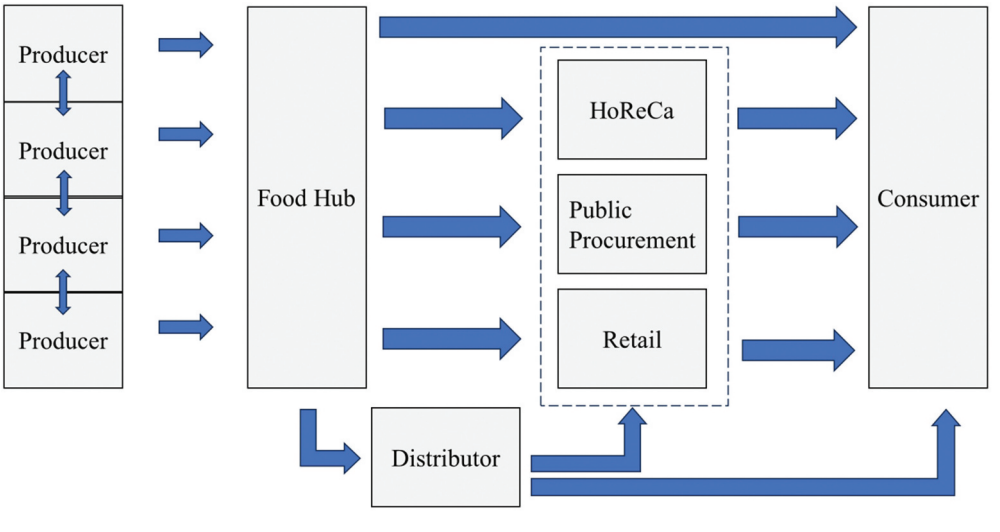


Figure 1. Food Hubs in SFSCs.

services, creating shared value and securing competitive advantage by leveraging socially embedded, difficult-to-replicate capabilities (Berti and Mulligan 2016; Tewari et al. 2018). By focusing on regional rather than hyperlocal SCs, they target SME producers and provide a viable alternative to CFSCs (Sgroi and Marino 2022; Tewari et al. 2018).

2.4. Capabilities of Food Hubs in SFSCs

FHs' capabilities can be grouped into three categories: operations management, producer support, and community engagement. While operational activities resemble those of conventional wholesalers and retailers, relational and mission-driven capabilities distinguish FHs in SFSCs. A full overview of the capabilities FHs have in SFSCs, and their sustainability indicators can be found in Table 1.

2.4.1. Operations management

FHs aggregate products, coordinate logistics, manage storage and distribution, set prices and engage in marketing communication of products (Barham et al. 2012; Fischer, Pirog, and Hamm 2015). They frequently own distribution infrastructure or form partnerships to overcome logistical barriers and solve distribution issues related to storage, warehousing and transportation that come with distributing perishable goods (Barham et al. 2012; Manikas, Malindretos, and Moschuris 2019; Sgroi and Marino 2022).

Many add value through packing or processing, reducing intermediaries and strengthening producer sustainability (Manikas, Malindretos, and Moschuris 2019). FHs also address food safety and quality standards for institutional buyers (Artoonian, Ross, and Shupp 2023; Matson and Thayer 2013), while innovative sales methods and broad networks help minimize food waste (Berti and Mulligan 2016; Hyland and Macken-Walsh 2022).

FHs help small and medium-sized producers overcome scale deficiencies by expanding local markets, creating demand for specialty products, and balancing supply and demand (Engelseth 2016; Hyland and Macken-Walsh 2022; Levidow and Psarikidou 2011; Pesci and Brinkley 2021; Rogers and Fraszczak 2014). Since many producers lack the power to negotiate with large buyers or meet strict requirements, FHs provide support in order management, invoicing, and price setting, thereby strengthening their market position (Artoonian, Ross, and Shupp 2023; Hyland and Macken-Walsh 2022; LeBlanc et al. 2014; Matson and Thayer 2013). By pooling multiple producers, FHs also generate economies of scale and act as mediators in negotiations with larger customers (Berti and Mulligan 2016; Manikas, Malindretos, and Moschuris 2019; Matson, Cook, and Sullins 2013).

2.4.2. Producer support

FHs differ from conventional intermediaries by also providing value addition support for producers, aiming to strengthen competitiveness in SFSCs (Fischer, Pirog, and Hamm 2015). They create shared value through production planning, capacity building through investments, and recruitment of new producers, while fostering collaboration and communication across the SC (Berti and Mulligan 2016; Conner et al. 2018; Matson and Thayer 2013). To access highly regulated markets such as public procurement, FHs coordinate production standards and support certification processes, either assisting producers

Table 1. Capabilities and sustainability of Food Hubs.

		Sustainability			
Category	Capabilities	Economic	Environm./ Health	Social	Sources
Operations management	Warehousing	X			Manikas, Malindretos, and Moschuris (2019); Sgroi and Marino (2022)
	Transportation from producers	X			Anggraeni, Handayati, and Novani (2022); Hyland and Macken-Walsh (2022)
	Transportation to customers	X			Berti and Mulligan (2016)
	Develop distribution channels		X		Barham et al. (2012)
	Aggregation	X			Barham et al. (2012); Berti and Mulligan (2016); Rogers and Fraszczak (2014)
	(Re-)Packing	X			Manikas, Malindretos, and Moschuris (2019)
	Processing	X			Manikas, Malindretos, and Moschuris (2019)
	Quality control		X		Artoonian, Ross, and Shupp (2023); Matson and Thayer 2013)
	Food safety		X		Artoonian, Ross, and Shupp (2023); Matson and Thayer (2013)
	Waste management		X		Berti and Mulligan (2016); Fischer, Pirog, and Hamm (2015); Manikas, Malindretos, and Moschuris (2019)
	Storage	X			Anggraeni, Handayati, and Novani (2022); Artoonian, Ross, and Shupp (2023); Berti and Mulligan (2016)
	Order management	X			Artoonian, Ross, and Shupp (2023); Hyland and Macken-Walsh (2022)
	Invoicing	X			Hyland and Macken-Walsh (2022); LeBlanc et al. (2014); Matson, Cook, and Sullins (2013)
	Price setting	X			(Berti and Mulligan)
	Develop sales methods	X			Berti and Mulligan (2016); Hyland and Macken-Walsh (2022)
	Marketing communications	X			Barham et al. (2012)
	Find new markets/ customers			X	Pesci and Brinkley (2021)
	Create demand for specialty products		X		Pesci and Brinkley (2021)
	Match supply and demand on local level			X	Christensen, Jablonski, and O'Hara (2019); Engelseth 2016; Matson, Cook, and Sullins (2013)
	Ensure consistent supply	X			Levidow and Psarikidou 2011; Pesci and Brinkley (2021)
	Negotiating with buyers	X			Berti and Mulligan (2016); Manikas, Malindretos, and Moschuris (2019); Matson, Cook, and Sullins (2013)
Producer support	Collaborative production planning	X			Berti and Mulligan (2016); Matson and Thayer (2013)
	Investment in producer capacities	X			Berti and Mulligan (2016); Matson and Thayer (2013)
	Development of innovative products	X			Manikas, Malindretos, and Moschuris (2019)
	Recruitment of producers	X			Berti and Mulligan (2016); Conner et al. (2018)
	Facilitate communication between producers			X	Manikas, Malindretos, and Moschuris (2019); Tewari et al. (2018)
	Coordinate production standards in SFSC		X		Matson, Cook, and Sullins (2013); Tewari et al. (2018)

(*Continued*)

Table 1. (Continued).

Category	Capabilities	Sustainability			Sources
		Economic	Environm./ Health	Social	
	Technical support for certification		X		Tewari et al. (2018)
Community engagement	Educational programming for customers			X	Cleveland et al. (2014); LeBlanc et al. (2014); Berti and Mulligan 2016; Tewari et al. 2018; Kawasaki et al. 2023)
	Facilitate communication between stakeholders			X	Manikas, Malindretos, and Moschuris (2019)
	Innovation brokers SFSC			X	Berti and Mulligan (2016); Hyland and Macken-Walsh (2022)
	Organize farm visits			X	Berti and Mulligan (2016); Cleveland et al. (2014)
	Donations to charity			X	Kawasaki et al. (2023); LeBlanc et al. (2014); Tewari et al. (2018).
	Institutional recombination (i.e. linking institutions and resources in new ways)			X	Tewari et al. (2018)
	Encourage training, innovation and new technologies			X	Berti and Mulligan (2016); Kao et al. (2024); Manikas, Malindretos, and Moschuris (2019)
	Focus on environmental practices		X		Manikas, Malindretos, and Moschuris (2019)
	Ensure healthy food for low-income and urban communities		X		Kao et al. (2024); Kawasaki et al. (2023); LeBlanc et al. (2014); Tewari et al. (2018).
	Create new jobs			X	Anggraeni, Handayati, and Novani (2022); Ilbery et al. (2004)

directly or taking responsibility after processing (Artoonian, Ross, and Shupp 2023; Cleveland et al. 2014; Matson, Cook, and Sullins 2013; Tewari et al. 2018).

2.4.3. Community engagement

Beyond economic and operational capabilities, FH's value-addition services also include the focus on community engagement (Fischer, Pirog, and Hamm 2015). Educating stakeholders, including consumers, is essential to increase awareness of SFSC benefits (Tewari et al. 2018). Transparency across stakeholders regarding production, processing, and logistics costs helps generate shared value (Barham et al. 2012; Berti and Mulligan 2016; Kao et al. 2024).

FHs often act as educators through initiatives such as farm visits (Cleveland et al. 2014; Hyland and Macken-Walsh 2022) and may foster institutional recombination by linking nonprofits, government agencies, and businesses, creating opportunities for funding and lowering entry barriers for producers (Tewari et al. 2018).

These activities, along with producer support, can stimulate market access, producer growth, and local job creation (Ilbery et al. 2004). Some FHs also engage in charitable food distribution to low-income communities (Kao et al. 2024), popularizing local food among children and supporting access to school food programs (Tewari et al. 2018). Such initiatives can enhance intergenerational knowledge, demand, and awareness of local food (Kao et al. 2024; Kawasaki et al. 2023; LeBlanc et al. 2014).

Finally, FHs contribute to healthier rural and urban diets by shortening time from harvest to sale, often with a stronger focus on environmental practices and animal welfare than conventional actors (Artoonian, Ross, and Shupp 2023).

3. Method

This study aims to increase the understanding of the capabilities of FHs and their sustainability indicators in SFSCs. This is done by conducting explorative case studies. The authors are using case study research as a research strategy that looks upon a phenomenon (the Food Hub) in its naturalistic context through a variety of data sources, to confront theory with the empirical world (Piekkari, Welch, and Paavilainen 2009). As previous research conducted within the field of FHs is limited, especially in a Swedish context, an inductive approach is appropriate (Yin 2018). The authors are furthermore based in Sweden; therefore, the selected companies are Swedish FHs.

3.1. Case selection

For a company to qualify as a FH, the key aspects that the researchers focused on included the company being an independent, profit-driven third-party organization, whose focus lies on the sale of exclusively locally or nationally produced food to customers. According to Yin (2018) the design of a multiple case study follows a similar logic, where each case must be carefully selected so that the individual case studies either: predict similar results (a literal replication) or predict contrasting results but for anticipable reasons (a theoretical replication) (Yin 2018). The goal of this study has not been to discover a generalization of the functions of FHs, rather than creating knowledge about the functions FHs fulfill in their specific context, which might lead to a replication in other FH-like organizations. Therewith, this study follows a theoretical replication.

With this in mind, the authors selected six different FHs (Table 2) that even though they follow the main definition of FH, they slightly differ from one and other: The FHs are located in different parts of Sweden; have different markets they focus on; and different prerequisites, such as geographical location, in which they operate (Table 2). FH A and B are operating mainly in the South of Sweden; FH C and D are mainly operating in the middle and west of Sweden; FH E is operating in the west of Sweden; and FH F is unique in the sense that it is operating on an island in the east of Sweden. Interestingly, the authors have not located any FHs in the Northen half of Sweden, which could be due to the subarctic climate with limited agricultural activities compared to the south.

3.2. Data collection

The authors started the research process by conducting a literature review which functions as the theoretical base of this paper. This theoretical base was used in the development of the interview guide in preparation for this study. However, while the interviewees were sent a concise semi-structured interview guide in advance of the interview, the interview was conducted as a semi-informal conversation. In this way, the interviewee was free to explain their knowledge and experience, without any predefined construct or theory. This is important, as the

Table 2. Presentation of case companies.

Case	FH A	FH B	FH C	FH D	FH E	FH F
No. employees	8	24	8	9	7	14
Established	2018	2008	2017	2015	2015	1995
Gross turnover	10,033,000 (2022) (−3,231,000)	76,485,000 (2022) (+408,000)	30,259,000 (2023) (+786,000)	29,285,000 (2023) (−48,000)	38,340,000 (2023) (+98,000)	111,954,000 (2022) (+5,033,000)
Main customer(s)	Consumer	HoReCa	HoReCa	HoReCa	Public institutions	HoReCa, Public institutions
Remaining customer(s)	Public institutions, HoReCa	Public institutions, Consumers	Public institutions	Public Institutions	HoReCa, Retail	Retail
Geographical location	South Sweden	South Sweden	East Sweden	West Sweden	West Sweden	East Sweden (Island)
No. interviews	4	4	2	1	2	3
Interviewees	1. CEO/Founder 2. Marketing Manager (2022 and 2023) 3. Sales	1. CEO/Founder 2. Sales 3.Warehouse/ Retail 4. Public Procurement	1. CEO/ Founder	1. CEO/ Founder 2. Sales/ Founder	1. CEO/ Founder 2. Co-owner/ Sales	1. Office Manager/ Sales 2. Retail sales 3. Warehouse

concepts/terms described by academic literature are not necessarily the terms that the interviewees would use in their description (Gioia, Corley, and Hamilton 2013). This could be due to different educational background, work experience, or simply different translation in the language, as the interviews were conducted in Swedish, while most academic literature is published in English. This created the opportunity for the discovery of new concepts, rather than the affirmation of existing concepts (Gioia, Corley, and Hamilton 2013).

The authors conducted 16 interviews with 15 employees in different positions within the organizations (see table 2). The interviews lasted between 30 and 90 minutes and took place at the FH's offices, except for five interviews which were done digitally. To ensure clarity and avoid any potential ambiguities, the researchers developed and tested a case study protocol before the interviews took place (Yin 2018). Furthermore, information about the research project, including the research questions as well as a short list over the main interview questions, were communicated to the case companies by e-mail in advance. At the beginning of every interview, the researcher conducting the interview explained the purpose of the study again and the interviewee had the opportunity to ask any questions they might have related to the study.

To strengthen the case study, and to be able to triangulate the data, the authors furthermore carried out observations of the companies' offices, warehouses and other operations; gathered documentation such as communications with suppliers and customers, as well as articles appearing in mass media; and consulted archival records obtaining official records such as annual financial reports. This was done to ensure that the case study's findings will be supported by more than a single source of evidence (Yin 2018).

3.3. Data analysis

The authors followed the principles of data collection as suggested by Yin (2018). They used multiple sources of evidence, leading to data triangulation (see 3.2). To ensure

reliability of the case studies, the authors reported and analyzed the data by forming structural databases in a qualitative data analysis software (Yin 2018). The collection of all data sources in one place can contribute to consistency and scientific accuracy (Yin 2018). To increase the construct validity of the study, the authors maintained a clear chain of evidence. In addition to carefully storing and analyzing the data in data analysis software, all interviews were recorded, with the permission of the interviewee, and later transcribed in accordance with the verbatim format. These transcriptions were thereafter sent back to the interviewee with the possibility to review.

The gathered data was structured and analyzed using Gioia inspired coding structures (Figure 2) (Gioia, Corley, and Hamilton 2013). In the lower levels of the Gioia coding scheme, the researchers have taken an inductive approach to analyze the data. While the overview of coding structure can be derived from the results of the earlier conducted SLR, by the same authors, taking a more deductive approach. Despite this, the authors goal was to increase knowledge with regard to the functions of FHs in a Swedish context, at the end of the study. Figure 2 shows an overview of the Gioia coding scheme with an overview over the first till third-order coding.

Once the data was structured, the authors studied the FH organizations first within-case to identify the unique functions and sustainability indicators of every FH and create a clear overview. Finally, the data were analyzed cross cases, trying to discover contrasting results amongst the cases (Yin 2018).

4. Results

The capabilities of FHs in SFSCs are diverse. They range from more traditional wholesale capabilities related to operations management including aggregation and distribution, to more value adding capabilities such as producer support, including direct investments, as well as community engagement involving the society.

As earlier mentioned, the studied FHs are mostly located in different parts of Sweden. They are similar in the sense that they fit the definition presented in the literature review, but different in terms of gross turnover and main customer markets. Interestingly all but two FHs (FH B and F) have been established between 2015 and 2018, in line with the increase of interest in SFSC in the scientific literature (Renkema and Hilletofth 2022). The FHs have been earlier presented in Table 2. This section will focus on a presentation of the different FH functions and their sustainability indicators, which will then be summarized in Table 3.

4.1. Operations management

Most of the studied FHs have operational capabilities related to operations management that are described in the literature. The concept of all the FHs is to aggregate the products and sell them in new ways to either customers such as HoReCa, public procurement, or consumers. To do so, FH B to F own or rent their own warehouse, where they have the capability to keep stock over an extended period. The exception is FH A, which does not keep a large stock of products in their warehouse yet:

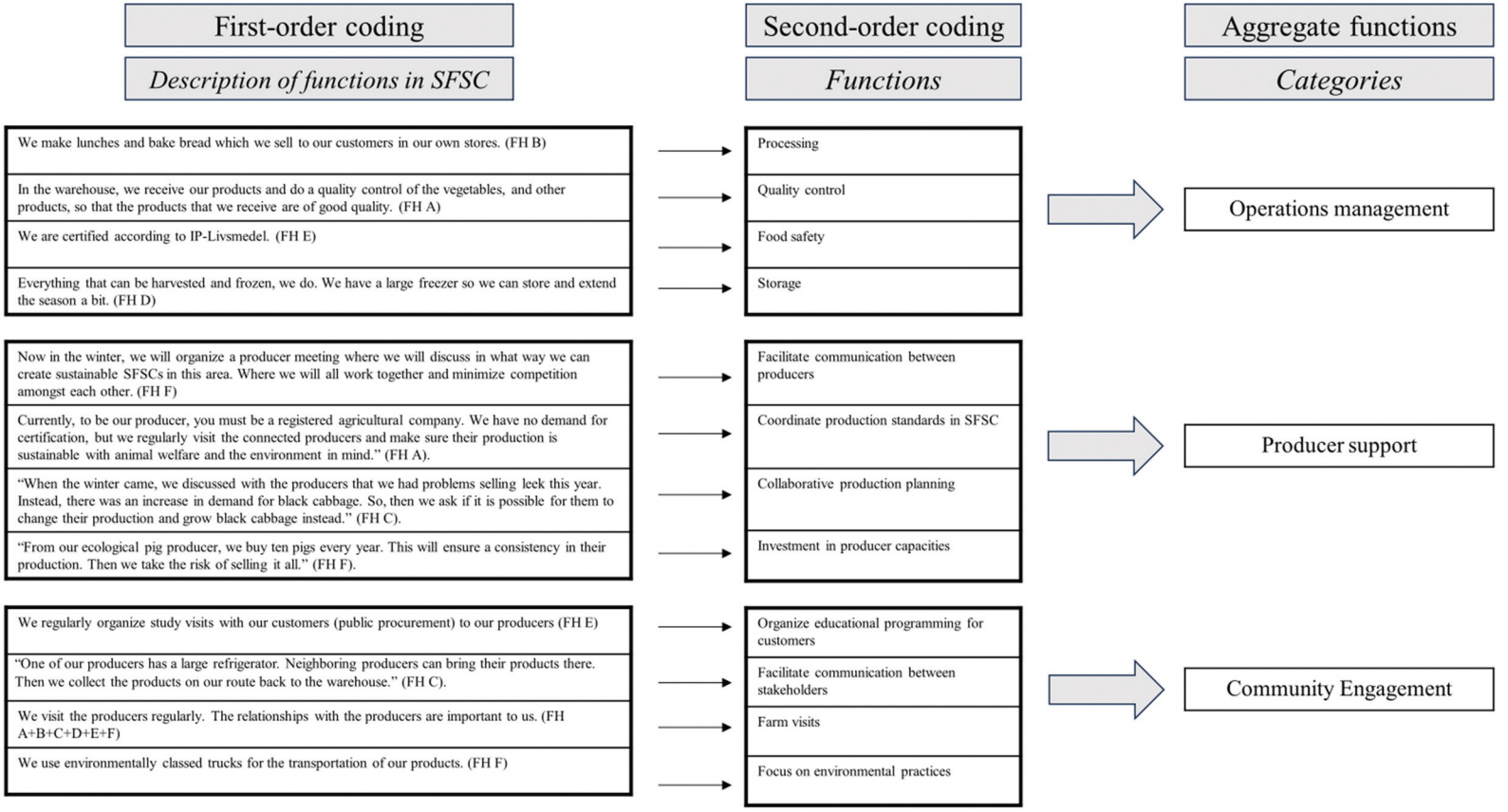


Figure 2. Illustration of the Gioia coding process.

> We do not have the space yet to store more products from different producers. We keep a very limited stock. Mostly we contact the producers after we receive an order, and they deliver the right amount to our warehouse.

Because of this, the FH has a lead time of four days from order to delivery.

As a vital part of the operations, all FHs organize the transportation not only to the customers and consumers but also arrange pick up from the producers to the warehouse, or even directly to the customers in case of large orders. The FHs either use existing logistic solutions (FH A, E and F) or develop their own (FH B, C, D, and F). FH A operates from a strategic location, in one of the major cities in Sweden, where it shares its warehouse with its collaboration partner, a national logistics company. Due to the national network of the existing logistics company, the consumers can be reached nationwide. FH B, C, and D, have its own logistics network with its own trucks. The trucks drive carefully planned routes where they combine pick-up from producers and deliver to customers as effectively as possible.

> We get the products from our producers in the morning, and we deliver it to the customers in the afternoon, on the way home. (FH B)

Like FH A, FH E shares its warehouse with a traditional wholesaler and outsources the transportation to an existing logistics firm. The location is strategic in the sense that the costs of pick-up and delivery decrease per unit when the trucks are shared and filled. FH F operates on an island, which is a touristic destination experiencing seasonal fluctuations. Therefore, they use both their own trucks, as well as outsource transportation to a logistics company.

> When we have the Easter Weekend, the demand on products increases enormously for only a few days. We need to scale up our transportation suddenly, but for a very short period. Then it is better for us to outsource to another company, rather than scaling up ourselves. (FH F)

While optimized transportation routes are a clear example of economic sustainability, it contributes together with the development of new distribution channels and the close collaboration with existing channels to decreased food miles leading to improved environmental sustainability.

Only FH B and C process and repack products that come in. At the warehouse of FH C, they process and repack meat before selling it to specific customers:

> At our facility, we dice meat, we hang meat, we make sausages, and we smoke meat, and we air dry meat and then we pack it and sell it to our customers. (FH C)

FH B has a more extensive processing facility with their own kitchen and retail shop attached to their warehouse. They offer not only products from producers but also make lunches and provide catering. By having the processing and repacking functions in house, the FH can decrease one otherwise necessary intermediary, keeping the SFSC as short as possible, leading to better economic sustainability for their own organization and the producers'.

The Swedish food industry is heavily regulated with strict production and hygiene rules that are to be followed, ensuring food safety. The FHs work only with certified producers and perform regular visits to the producers to maintain trust and good relationships.

Table 3. Capabilities and sustainability of Food Hubs case studies.

		Sustainability								
Category	Capabilities	Economic	Environm./ Health	Social	FH A	FH B	FH C	FH D	FH E	FH F
Operations management	Warehousing	X			–	X	X	X	X	X
	Transportation from producers	X			X	X	X	X	X	X
	Transportation to customers	X			X	X	X	X	X	X
	Develop distribution channels		X		X	X	X	X	X	X
	Aggregation	X			X	X	X	X	X	X
	(Re-)Packing	X			-	X	X	–	–	–
	Processing	X			-	X	X	–	–	–
	Quality control		X		X	X	X	X	X	X
	Food safety		X		X	X	X	X	X	X
	Waste management		X		X	X	X	X	X	X
	Storage	X			-	X	X	X	X	X
	Order management	X			X	X	X	X	X	X
	Invoicing	X			X	X	X	X	X	X
	Price setting	X			X	X	X	X	X	X
	Develop sales methods	X			X	X	X	X	X	X
	Marketing communications	X			X	X	X	X	X	X
	Find new markets/ customers			X	X	X	X	X	X	X
	Create demand for specialty products		X		X	X	X	X	X	X
	Match supply and demand on local level			X	X	X	X	X	X	X
	Ensure consistent supply	X			X	X	X	X	X	X
	Negotiating with buyers	X			X	X	X	X	X	X
Producer support	Collaborative production planning	X			–	X	X	X	X	X
	Investment in producer capacities	X			-	X	–	–	–	X
	Development of innovative products	X			-	–	–	–	–	–
	Recruitment of producers	X			X	X	X	X	X	X
	Facilitate communication between producers			X	X	X	X	X	X	X
	Coordinate production standards in SFSC		X		X	X	X	X	X	X
	Technical support for certification		X		-	–	X	X	–	–
Community engagement	Educational programming for customers			X	–	–	–	–	–	–
	Facilitate communication between stakeholders			X	X	X	X	X	X	X
	Innovation brokers SFSC			X	X	–	–	–	–	–
	Organize farm visits			X	–	–	–	–	X	–
	Donations to charity			X	–	–	–	–	–	–

(*Continued*)

Table 3. (Continued).

Category	Capabilities	Sustainability			FH A	FH B	FH C	FH D	FH E	FH F
		Economic	Environm./ Health	Social						
	Institutional recombination (i.e. linking institutions and resources in new ways)			X	X	–	–	–	–	–
	Encourage training, innovation and new technologies			X	–	–	–	–	–	–
	Focus on environmental practices		X		X	X	X	X	X	X
	Ensure healthy food for low-income and urban communities		X		-	–	–	–	–	–
	Create new jobs			X	–	–	–	–	–	–

Moreover, products are checked when they are delivered to the warehouse and before sending it to customers and consumers.

> We are daily out there, at the producers, and see their companies. We maintain close relations with all our producers and know exactly how their production is going. We see and feel the products when they arrive at our warehouse. (FH B)

If there are any products coming closer to their expiration date, the FHs either make a more active effort to sell in these products to customers, or in the case of FH B, these products can be processed further into ready-made meals, being sold at their own shop, extending the shelf life of the products. These functions are closely related to environmental and health sustainability.

All six FHs originated from the ambition to empower local producers, increase the production and sales of locally produced food, and create an overall more sustainable SFSC. The FHs believe that the producers' main concern should be to focus on their production, as they are the ones who are experts in that. The FH in its turn will take care of the financial administration connected to the sales such as price setting and invoicing:

> Our producers only send the bills for their production to us. We in our turn, invoice our customers separately and take out a commission for our mediating role. In this way, our producer does not need to book times with 15 different customers and send separate invoices to all of them. (FH F)

The FHs furthermore take care of marketing and sales functions that include developing sales methods, marketing communications, finding new markets and creating demand for specialty products:

> We follow the trends in the markets, and we can ask our producers to grow for example, edible flowers, which we then buy, and sell to our customers. (FH C)

By finding new markets, and creating markets for specialty products, the FHs can encourage development of rural areas, contributing to both social as well as environmental sustainability of the rural areas (Deller, Lamie, and Stickel 2017).

One of the challenges that small and medium sized producers face is their ability to ensure consistent supply year-round, and to find customers that are willing to be agile as a consequence (Artoonian, Ross, and Shupp 2023). The FHs work actively for their producers to match the supply and demand to ensure the consistent supply. And with disruptions in the market, they serve as mediators in the negotiations with larger buyers:

> We are bound by contracts in our collaboration with government institutions. We can only change sale prices on fixed times in the year. However, in times of crisis, for example when the electricity prices went up, local producers were forced to increase their prices. We try to negotiate these new prices with the city council outside of the fixed times. (FH B)

4.2. Producer support

Since FHs recognizes producers as their strategic partners in SFSCs, producer support is a vital part of the FH's relational capabilities. All FHs actively recruit small and medium-sized producers in the regions they operate in. Through different activities, and networking, the FHs not only maintain the relationships with the producers, but also facilitate communication amongst the different producers:

> Now in the winter, we will organize a producer meeting where we will discuss in what way we can create sustainable SFSCs in this area. Where we will all work together and minimize competition amongst each other. (FH F)

These actions do not only minimize competition between the different producers, with the negative economic consequences that it might give, but it enables the development of inclusive and social sustainable SFSCs.

All FHs show a strong awareness of the environment and health sustainability indicators by focusing on the coordination of production standards in the SFSC:

> Currently, in order to be our producer, you have to be a registered agricultural company. We have no demand for certification, but we regularly visit the connected producers and make sure their production is sustainable with animal welfare and the environment in mind. (FH A)

To match supply and demand, and ensure consistent supply as mentioned under operations management, FH B to F collaborate closely with producers in the planning of their production:

> When the winter came, we discussed with the producers that we had problems selling leek this year. Instead, there was an increase in demand for black cabbage. So, then we ask if it is possible for them to change their production and grow black cabbage instead. (FH C)

Sometimes the FH recognizes a demand in the market that is not being met, and then some of the FHs invest in the producer's capacity to meet the demand:

> From our ecological pig producer, we buy ten pigs every year. This will ensure consistency in their production. Then we take the risk of selling it all. (FH F)

While the FHs support the producers in different ways, the data did not show any involvement in the development of innovative products, nor do the FHs provide technical support for certification.

4.3. Community engagement

To increase shared value that will lead to increased sustainability in the SFSCs, some of the FH's functions should focus on identifying and expanding the connections between societal and economic progress (Porter and Kramer 2011). The literature recognizes several capabilities related to community engagement in which FHs have the possibility to increase mainly social sustainability, but also environmental sustainability.

While analyzing the case study data the FHs appear to have limited involvement in community engagement. Only FH E organizes educational programming for customers, and none of them donate to charities. It was furthermore not clear in what way they encourage training, innovation and new technologies in the agricultural sector. Finally, with local food products still having the image of being more expensive, they do not directly ensure healthy food for low-income and urban communities:

> We mainly deliver to restaurants, and they are not very price sensitive. I will always work hard to sell in our island's local products, even if it means it is a bit more expensive for the customers. (FH F)

Only one of the FHs studied, FH A, was a clear innovation broker in SFSC. This company developed themselves as an innovative retailer with their own digital platform, where consumers can order 'directly from the producers'. They are trying to create awareness in the market for the existing producers and their products, while offering competitive prices compared to traditional retailers. By creating trust, awareness and relationship between the producers and the consumers FH A has a clear focus on social sustainability in the SFSC.

As mentioned under producer support, the relationships between the producers and the FHs are based on trust and communication. They work to not only increase the relationships between FH and individual producers, but also encourage and enable collaboration between the different producers:

> One of our producers has a large refrigerator. Neighboring producers can bring their products there. Then we collect the products on our route back to the warehouse. (FH C)

This increases the overall accessibility of small- and medium-sized producers to operate in SFSCs and benefit the community, contributing to social sustainability. This might instigate a long-term effect of increased profit for producers, leading them to grow, which in turn might create job opportunities. It was, however, not specifically clear that the actions of the studied FHs directly create new jobs.

As mediators in the SFSCs, FHs can link institutions and resources in new ways by means of institutional recombination (Tewari et al. 2018). In the case of FH A, they actively work together with different financial institutions and react to public investment calls, to secure finance for their own organization and the producers connected to them. While the literature recognizes the FH's role in donating to charities such as food banks, an aspect of social sustainability, this did not appear from the FHs in this study.

Finally, all FHs have a clear focus on environmental sustainability related to community engagement. FH F emphasized during the interview the use of environmentally classed trucks they are using. The FHs in the study do not require their producers to be organically, or otherwise, certified, as the process is an expensive and complicated one. They do,

however, encourage their producers to produce food in environmentally friendly ways, and with a focus on animal welfare. FH B to F are furthermore certified according to the Swedish IP-standard.[1]

5. Discussion

The findings indicate that the investigated FHs perform most of the operations management functions commonly identified in the literature, with only minor variation across cases. These capabilities represent valuable resources that strengthen both their own profitability and that of their producers by shortening the SC and reducing reliance on external intermediaries. By combining operational efficiency with producer support and community engagement, FHs enhance both economic competitiveness and the broader sustainability of SFSCs. Their ability to mobilize relational and socially embedded capabilities such as trust, collaboration, and engagement, differentiates them from conventional intermediaries and creates long-term value for producers, consumers, and communities thereby supporting competitive advantage in SFSCs.

The sustainability indicators linked to operations management are primarily of economic nature, aimed at reducing costs for the FH. Although quality control, food safety, and waste management also reflect environmental and health dimensions. To reduce food miles and strengthen environmental sustainability, FHs apply different strategies by either optimizing transport routes by combining producer pick-ups with customer deliveries or partnering with existing logistics firms. Such operational efficiencies represent valuable resources that enhance competitiveness when tailored to the constraints of SFSCs (Barney 1991; Christopher 2023).

Notably the relational resources play a crucial role in differentiating FHs from conventional intermediaries. Trust and collaboration underpin SFSCs, and all FHs actively invest in relationships with and among producers. Only FH B and FH F extend this further by investing directly in producer capacities. Their ability to do so reflects their higher turnover and larger workforce, suggesting stronger financial resources that enable greater capability development. Trust and collaboration through support and investments represent unique capabilities that not only strengthen FH and SFSC stakeholder competitiveness but also secure long-term relational advantages for the FHs.

The most notable divergence between the FHs studied here and those in prior literature lies in community engagement. While Swedish FHs facilitate stakeholder communication, promote environmental practices, and engage in institutional recombination, they do not prioritize educational initiatives, training, charitable donations, or programs supporting low-income consumers. This limited scope may reflect their for-profit orientation, which prioritizes economically valuable capabilities over socially embedded, community-driven capabilities. Literature suggests that FHs engaging the community more deeply in their operations achieve higher economies of scale, which may enhance their competitive advantage and economic sustainability (Kao et al. 2024).

6. Conclusions, implications and further research

To increase the competitive advantage of SFSCs, this paper investigated the capabilities of FHs in SFSCs and their impact on sustainability indicators. The outcome presented three

categories namely operations management, producer support and community engagement. While on a holistic level there does not seem to be a large difference between the studied FHs, the table clearly shows the differences with FHs previously studied in other countries in the literature.

The studied FHs create value in the SFSCs by their large focus on economic sustainability for their organization as well as the small- and medium-sized producers. As well as their focus on the production and sales of healthy food that is produced in an environmentally conscious way.

Compared to FHs documented in the literature, these FHs show limited involvement in community engagement, which may undermine their long-term profitability. Strengthening community engagement, through consumer education and by encouraging producers to adopt innovative production practices, could enhance demand for local products while simultaneously improving and expanding local food production.

Several avenues emerge for future research on FHs. First, the role of retailers as customers warrants closer examination. Retailers increasingly adopt omni-channel strategies to reach consumers, integrating online and offline channels. Understanding how these strategies influence FH operations, capabilities, and value creation could provide insights into the alignment between FHs and downstream market demands.

Second, current research largely examines FH capabilities from a FH-centric perspective, without incorporating the views of other stakeholders. Future studies should explore stakeholder engagement and relational management, considering producers, consumers, retailers, and other partners. This would clarify how FHs leverage their relational and operational capabilities to coordinate across diverse actors, create shared value, and strengthen their competitive advantage in SFSC.

Finally, comparative studies on collaborations among FHs and between FHs and other SC stakeholders could illuminate how inter-organizational networks contribute to the development of capabilities, both operational and relational, and identify the mechanisms through which these capabilities support sustainable performance and market differentiation.

While this paper identifies a range of sustainability indicators associated with FH capabilities, it does not provide empirical measurement of these indicators. Consequently, it remains unclear whether they translate into tangible improvements in the sustainability of SFSCs. Future research should therefore develop models to assess the concrete outcomes of these capabilities in relation to sustainability performance. In particular, quantitative approaches are needed to evaluate such outcomes, for example through the application of life cycle assessment (LCA) methods to capture environmental sustainability impacts.

The capabilities that are presented are quite similar among the different studied FHs. However, company presentations and quotes indicate different business models and growth strategies. Future research could look at the different sustainable business models of these FHs, and compare it based on customer base and geographical location.

Note

1. The IP standard – a Nordic standard for more sustainable production of food and flowers in the areas of food safety, animal care, environmental responsibility and fair working conditions. https://www.sigill.se/.

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References

Allen P, FitzSimmons M, Goodman M and Warner K. (2003). Shifting plates in the agrifood landscape: the tectonics of alternative agrifood initiatives in California. Journal of Rural Studies, 19 (1): 61–75. https://doi.org/10.1016/S0743-0167(02)00047-5.

Anggraeni, E. W., Y. Handayati, and S. Novani. 2022. "Improving Local Food Systems Through the Coordination of Agriculture Supply Chain Actors." *Sustainability (Switzerland)* 14 (6). https://doi.org/10.3390/su14063281.

Artoonian, T. A., R. B. Ross, and R. S. Shupp. 2023. "Identification and Assessment of Supply Chain Risks: The Case of Food Hubs." *Agribusiness* 41: 1–22. https://doi.org/10.1002/agr.21877.

Barham, J., D. Tropp, K. Enterline, J. Farbman, J. Fisk, and S. Kiraly. 2012. *Regional Food Hub Resource Guide*. U.S. Dept. of Agriculture, Agricultural Marketing Service.

Barney, J. 1991. "Firm Resources and Sustained Competitive Advantage." *Journal of Management* 17: 99–120. https://doi.org/10.1177/014920639101700108.

Berti, G., and C. Mulligan. 2016. "Competitiveness of Small Farms and Innovative Food Supply Chains: The Role of Food Hubs in Creating Sustainable Regional and Local Food Systems." *Sustainability (United States)* 8 (7). https://doi.org/10.3390/su8070616.

Bloom, J. D., and C. C. Hinrichs. 2011. "Moving Local Food Through Conventional Food System Infrastructure: Value Chain Framework Comparisons and Insights." *Renewable Agriculture and Food Systems* 26 (1): 13–23. https://doi.org/10.1017/S1742170510000384.

Born, B., and M. Purcell. 2006. "Avoiding the Local Trap: Scale and Food Systems in Planning Research." *Journal of Planning and Educational Research* 26: 195–207. https://doi.org/10.1177/0739456X06291389.

Bosona, T. G., and G. Gebresenbet. 2011. "Cluster Building and Logistics Network Integration of Local Food Supply Chain." *Biosystems Engineering* 108 (4): 293–302. https://doi.org/10.1016/j.biosystemseng.2011.01.001.

Brunori, G., F. Galli, D. Barjolle, R. van Broekhuizen, L. Colombo, M. Giampietro, J. Kirwan, et al. 2016. "Are Local Food Chains More Sustainable Than Global Food Chains? Considerations for Assessment." *Sustainability (Switzerland)* 8 (5): 1–27. https://doi.org/10.3390/su8050449.

Charatsari C, Lioutas E D and De Rosa M. (2024). Going Short and Going Digital: How Do Consumers View the Impacts of Digitalizing Short Food Supply Chains?. Sustainability, 16 (24): 11241 https://doi.org/10.3390/su162411241.

Chiffoleau Y and Dourian T. (2020). Sustainable Food Supply Chains: Is Shortening the Answer? A Literature Review for a Research and Innovation Agenda. Sustainability, 12 (23): 9831 https://doi.org/10.3390/su12239831.

Christensen, L. O., B. B. R. Jablonski, and J. K. O'Hara. 2019. "School Districts and Their Local Food Supply Chains." *Renewable Agriculture and Food Systems* 34 (3): 207–215. https://doi.org/10.1017/S1742170517000540.

Christopher, M. 2023. *Logistics and Supply Chain Management*. 6th ed. Harlow: Pearson Education.

Cleveland, D. A., N. M. Müller, A. C. Tranovich, D. N. Mazaroli, and K. Hinson. 2014. "Local Food Hubs for Alternative Food Systems: A Case Study from Santa Barbara County, California." *Journal of Rural Studies* 35: 26–36. https://doi.org/10.1016/j.jrurstud.2014.03.008.

Conner, D. S., K. Sims, R. Berkfield, and H. Harrington. 2018. "Do Farmers and Other Suppliers Benefit from Sales to Food Hubs? Evidence from Vermont." *Journal of Hunger & Environmental Nutrition* 13 (4): 507–516. https://doi.org/10.1080/19320248.2017.1378602.

Deller, S. C., D. Lamie, and M. Stickel. 2017. "Local Foods Systems and Community Economic Development." *Community Development* 48 (5): 612–638. https://doi.org/10.1080/15575330.2017.1373136.

Dimitri, C., and K. Gardner. 2019. "Farmer Use of Intermediated Market Channels: A Review." *Renewable Agriculture and Food Systems* 34 (3): 181–197. https://doi.org/10.1017/S1742170518000182.

Engelseth, P. 2016. "Developing Exchange in Short Local Foods Supply Chains." *International Journal on Food System Dynamics* 7 (3): 229–242. https://doi.org/10.18461/ijfsd.v7i3.734.

Falasco, S., G. Ferla, P. Caputo, B. Mura, B. Borrello, M. Marson, V. León Bravo, F., and Caniato. 2025. "The impacts of food chains: are alternative food networks sustainable?" *Clean Technologies and Environmental Policy*. https://doi.org/10.1007/s10098-025-03262-1 .

Fischer, M., R. Pirog, and M. W. Hamm. 2015. "Food Hubs: Definitions, Expectations, and Realities." *Journal of Hunger & Environmental Nutrition* 10 (1): 92–99. https://doi.org/10.1080/19320248.2015.1004215.

Galli, F., and G. Brunori. 2013. *Short Food Supply Chains as Drivers of Sustainable Development Evidence Document*. Laboratoria di studi rurali Sismondi.

Gioia, D. A., K. G. Corley, and A. L. Hamilton. 2013. "Seeking Qualitative Rigor in Inductive Research: Notes on the Gioia Methodology." *Organizational Research Methods* 16 (1): 15–31. https://doi.org/10.1177/1094428112452151.

Gold, S., S. Seuring, and P. Beske. 2010. "Sustainable Supply Chain Management and Inter-Organizational Resources: A Literature Review." *Corporate Social Responsibility & Environmental Management* 17: 230–245. https://doi.org/10.1002/csr.207.

Hinrichs, C. C. 2003. "The Practice and Politics of Food System Localization." *Journal of Rural Studies* 19:33–45. https://doi.org/10.1016/S0743-0167(02)00040-2.

Hyland, J. J., and Á. Macken-Walsh. 2022. "Multi-Actor Social Networks: A Social Practice Approach to Understanding Food Hubs." *Sustainability (Switzerland)* 14 (3). https://doi.org/10.3390/su14031894.

Ilbery, B., D. Maye, M. Kneafsey, T. Jenkins, and C. Walkley. 2004. "Forecasting Food Supply Chain Developments in Lagging Rural Regions: Evidence from the UK." *Journal of Rural Studies* 20 (3): 331–344. https://doi.org/10.1016/j.jrurstud.2003.09.001.

Kao, J., G. Woodward-Lopez, M. I. Rangel, and A. Hicks. 2024. "Lessons Learned from a Food Hub-Based Retail Intervention That Expanded Reach and Sales of Fresh Fruits and Vegetables in Low-Income, Low-Access Neighborhoods." *Journal of Hunger & Environmental Nutrition* 19: 1145–1164. https://doi.org/10.1080/19320248.2023.2285916.

Kawasaki, N., T. Washio, K. Nakamura, and K. I. Nagahama. 2023. "Maintaining Agricultural Production by Building Local Distribution Systems in the Northern Area of Japan." *The Land* 12 (2). https://doi.org/10.3390/land12020320.

Kim M, Song J and Triche J. (2015). Toward an integrated framework for innovation in service: A resource-based view and dynamic capabilities approach. Inf Syst Front, 17 (3): 533–546. https://doi.org/10.1007/s10796-014-9505-6.

Landon, A., and M. Rosol. 2025. "Overcoming the Local Trap Through Inclusive and Multi-Scalar Food Systems." *Local Environment* 30: 923–943. https://doi.org/10.1080/13549839.2025.2450492.

LeBlanc, J., D. Conner, G. McRae, and H. Darby. 2014. "Building Resilience in Nonprofit Food Hubs." *Journal of Agriculture, Food, Systems, and Community Development* 4 (3): 121–135. https://doi.org/10.5304/jafscd.2014.043.005.

Levidow, L., and K. Psarikidou. 2011. "Food Relocalization for Environmental Sustainability in Cumbria." *Sustainability* 3 (4): 692–719. https://doi.org/10.3390/su3040692.

Malak-Rawlikowska, A., E. Majewski, A. Was, S. O. Borgen, P. Csillag, M. Donati, R. Freeman, et al. 2019. "Measuring the Economic, Environmental, and Social Sustainability of Short Food Supply Chains." *Sustainability (Switzerland)* 11 (15). https://doi.org/10.3390/su11154004.

Manikas, I., G. Malindretos, and S. Moschuris. 2019. "A Community-Based Agro-Food Hub Model for Sustainable Farming." *Sustainability (Switzerland)* 11 (4). https://doi.org/10.3390/su11041017.

Matson, J., C. Cook, and M. Sullins. 2013. *The Role of Food Hubs in Local Food Marketing*. USDA Rural Development. Service Report 73, 56.

Matson, J., and J. Thayer. 2013. "The Role of Food Hubs in Food Supply Chains." *Journal of Agriculture, Food, Systems, and Community Development* 3 (4): 1–5. https://doi.org/10.5304/jafscd.2013.034.004.

Mejía G and García-Díaz C. (2018). Market-level effects of firm-level adaptation and intermediation in networked markets of fresh foods: A case study in Colombia. Agricultural Systems, 160: 132–142. https://doi.org/10.1016/j.agsy.2017.06.003.

Mustapa M Adzran and Kallas Z. (2025). Meta-Analysis of Consumer Willingness to Pay for Short Food Supply Chain Products. Global Challenges, 9(3), https://doi.org/10.1002/gch2.202400154.

Nordmark, I., D. Ljungberg, G. Gebresenbet, T. Bosona, and R. Jüriado. 2012. "Integrated Logistics Network for the Supply Chain of Locally Produced Food, Part II: Assessment of e-Trade, Economic Benefit and Environmental Impact." *Journal of Service Science & Management* 5 (3): 249–262. https://doi.org/10.4236/jssm.2012.53030.

Pesci, S., and C. Brinkley. 2021. "Can a Farm-to-Table Restaurant Bring About Change in the Food System?: A Case Study of Chez Panisse." *Food, Culture, and Society* 25 (5): 997–1018. https://doi.org/10.1080/15528014.2021.1948754.

Piekkari, R., C. Welch, and E. Paavilainen. 2009. "The Case Study as Disciplinary Convention: Evidence from International Business Journals." *Organizational Research Methods* 12 (3): 567–589. https://doi.org/10.1177/1094428108319905.

Porter, M. E., and M. R. Kramer. 2011. "How to Reinvent Capitalism—and Unleash a Wave of Innovation and Growth." *Harvard Business Review* Jan-Feb: 1–17 Jan-Feb.

Remvig, S., and P. Hansson. 2022. *Prisökningar på lantbrukets insatsvaror – utmaningar och åtgärder*. Uppsala, Sweden: SLU Future Food.

Renkema, M., and P. Hilletofth. 2022. "Intermediate Short Food Supply Chains: A Systematic Review." *British Food Journal* 124 (13): 541–558. https://doi.org/10.1108/BFJ-06-2022-0463.

Rogers, J., and M. Fraszczak. 2014. "'Like the Stem Connecting the Cherry to the Tree': The Uncomfortable Place of Intermediaries in a Local Organic Food Chain." *Sociologia Ruralis* 54 (3): 321–340. https://doi.org/10.1111/soru.12041.

Sage, C. 2003. "Social Embeddedness and Relations of Regard." *Journal of Rural Studies* 19: 47–60. https://doi.org/10.1016/s0743-0167(02)00044-x.

Schreiber K, Soubry B, Dove-McFalls C and MacDonald G K. (2023). Untangling the role of social relationships for overcoming challenges in local food systems: a case study of farmers in Québec, Canada. Agric Hum Values, 40(1): 141–156. https://doi.org/10.1007/s10460-022-10343-0.

Sgroi, F., and G. Marino. 2022. "Environmental and Digital Innovation in Food: The Role of Digital Food Hubs in the Creation of Sustainable Local Agri-Food Systems." *Science of the Total Environment* 810 (2022): 152257. https://doi.org/10.1016/j.scitotenv.2021.152257.

Stevenson, G. W., K. Clancy, R. King, L. Lev, M. Ostrom, and S. Smith. 2011. "Midscale Food Value Chains: An Introduction." *Journal of Agriculture, Food, Systems, and Community Development* 1 (4): 27–34. https://doi.org/10.5304/jafscd.2011.014.007.

Tewari, M., S. Kelmenson, A. Guinn, G. Cumming, and R. Colloredo-Mansfeld. 2018. "Mission-Driven Intermediaries as Anchors of the Middle Ground in the American Food System: Evidence from Warrenton, NC." *Culture, Agriculture, Food and Environment* 40 (2): 114–123. https://doi.org/10.1111/cuag.12175.

Trivette, S. A. 2019. "The Importance of Food Retailers: Applying Network Analysis Techniques to the Study of Local Food Systems." *Agriculture and Human Values* 36 (1): 77–90. https://doi.org/10.1007/s10460-018-9885-1.

Vittersø, G., H. Torjusen, K. Laitala, B. Tocco, B. Biasini, P. Csillag, M. D. de Labarre, et al. 2019. "Short Food Supply Chains and Their Contributions to Sustainability: Participants' Views and Perceptions from 12 European Cases." *Sustainability (Switzerland)* 11 (17): 4800. https://doi.org/10.3390/su11174800.

Yin, R. K. 2018. *Case Study Research and Applications: Design and Methods*. Thousand Oaks, CA: Sage Publications, Inc.